WORKED SOLUTION

STEP III 2003, Question 14

Probability and Statistics • Probability Generating Functions

STEP III 2003, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q14

The key idea

The probability generating function (pgf) of a sum of independent scores is the product of their pgfs. So a question about *which dice give the same distribution of totals* becomes a question about *how a polynomial can be split into two factors*. Factorise everything into irreducible pieces, and the possible relabellings can be read off by sharing those pieces out between the two dice.

Solution

First part

Let X be the score on a standard fair die. Each of $1,\dots,6$ has probability $\frac{1}{6}$, so

$$G_X(t) = \mathrm{E}(t^X) = \tfrac{1}{6}(t + t^2 + t^3 + t^4 + t^5 + t^6) = \tfrac{1}{6}\,t\,(1 + t + t^2 + t^3 + t^4 + t^5).$$

To factorise the bracket, pair up terms and then use a difference of cubes:

$$1 + t + t^2 + t^3 + t^4 + t^5 = (1 + t) + t^2(1 + t) + t^4(1 + t) = (1 + t)(1 + t^2 + t^4),$$
$$1 + t^2 + t^4 = (1 + t^2)^2 - t^2 = (1 + t + t^2)(1 - t + t^2).$$

Hence

$$G_X(t) = \tfrac{1}{6}\,t\,(1 + t)(1 - t + t^2)(1 + t + t^2).$$

If X_1 and X_2 are the scores on two dice rolled independently, then t^{X_1} and t^{X_2} are independent, so $\mathrm{E}(t^{X_1+X_2}) = \mathrm{E}(t^{X_1})\mathrm{E}(t^{X_2})$ and

$$G_{X_1+X_2}(t) = G_X(t)^2 = \frac{1}{36}\,t^2(1 + t)^2(1 - t + t^2)^2(1 + t + t^2)^2,$$

which is the required form.

Second part

Let Y and Z be the scores on the dice labelled $1,2,2,3,3,4$ and $1,3,4,5,6,8$. Counting faces,

$$G_Y(t) = \tfrac{1}{6}(t + 2t^2 + 2t^3 + t^4),$$
$$G_Z(t) = \tfrac{1}{6}(t + t^3 + t^4 + t^5 + t^6 + t^8).$$

For Y: $1 + 2t + 2t^2 + t^3 = (1 + t)(1 + t + t^2)$, which you can check by expanding. So

$$G_Y(t) = \tfrac{1}{6}\, t\,(1 + t)(1 + t + t^2).$$

For Z, the useful products from the first part are $(1 + t)(1 - t + t^2) = 1 + t^3$ and $(1 + t + t^2)(1 - t + t^2) = 1 + t^2 + t^4$. Then

$$(1 + t^3)(1 + t^2 + t^4) = 1 + t^2 + t^3 + t^4 + t^5 + t^7,$$

and multiplying by t gives exactly $t + t^3 + t^4 + t^5 + t^6 + t^8$. Therefore

$$G_Z(t) = \tfrac{1}{6}\, t\,(1 + t)(1 + t + t^2)(1 - t + t^2)^2.$$

Rolling independently,

$$G_{Y+Z}(t) = G_Y(t)G_Z(t) = \frac{1}{36}\, t^2(1 + t)^2(1 - t + t^2)^2(1 + t + t^2)^2 = G_{X_1+X_2}(t).$$

Two pgfs that are equal as polynomials have equal coefficients, and the coefficient of t^n is the probability that the total is n. So every total has the same probability with these dice as with two standard dice.

Final part

A standard four-faced die has pgf

$$\tfrac{1}{4}(t + t^2 + t^3 + t^4) = \tfrac{1}{4}\, t\,(1 + t)(1 + t^2),$$

so the target pgf for the total of two of them is

$$\frac{1}{16}\, t^2(1 + t)^2(1 + t^2)^2.$$

Suppose the two new dice have pgfs $\frac{1}{4}f(t)$ and $\frac{1}{4}g(t)$. The coefficient of t^n in f is the number of faces labelled n, so f and g are polynomials with non-negative integer coefficients and $f(1) = g(1) = 4$. Since every label is a positive integer, $f(0) = g(0) = 0$. We need

$$f(t)\, g(t) = t^2(1 + t)^2(1 + t^2)^2.$$

Ask yourself Why must f be built from the same factors t, $1 + t$, $1 + t^2$, and nothing else?

The factors t and $1+t$ are linear, and $1+t^2$ has no real roots, so all three are irreducible over the real numbers. Polynomials factorise uniquely into irreducibles (up to constant multiples), so any factor of the right-hand side is

$$f(t) = c\,t^k(1+t)^a(1+t^2)^b, \qquad g(t) = \tfrac{1}{c}\,t^{2-k}(1+t)^{2-a}(1+t^2)^{2-b},$$

with $0 \le k, a, b \le 2$ and a constant $c > 0$.

Now we use the conditions on f and g one at a time.

- *The constant c.* The lowest-degree coefficient of f is c, and that of g is $1/c$. Both are counts of faces, so both are positive integers, which forces $c = 1$.
- *The power of t.* $f(0) = 0$ needs $k \ge 1$, and $g(0) = 0$ needs $2 - k \ge 1$. So $k = 1$: each die has exactly one factor t.
- *Four faces.* $f(1) = 2^a \cdot 2^b = 4$ gives $a + b = 2$, and then $g(1) = 2^{2-a}2^{2-b} = 4$ automatically.

So (a, b) is $(2,0)$, $(1,1)$ or $(0,2)$, with g taking the complementary choice. Since swapping the two dice gives the same pair, there are two cases.

- $(a, b) = (1,1)$ for both: $f = g = t(1+t)(1+t^2) = t + t^2 + t^3 + t^4$, the standard pair $1,2,3,4$ and $1,2,3,4$.
- $(a, b) = (2,0)$ and $(0,2)$:

  $$f(t) = t(1+t)^2 = t + 2t^2 + t^3, \qquad g(t) = t(1+t^2)^2 = t + 2t^3 + t^5,$$

  which are the dice labelled $1,2,2,3$ and $1,3,3,5$.

Both polynomials in each case have non-negative integer coefficients summing to 4, so both really are dice, and by construction they give the right pgf.

Hence the only labellings are the standard pair, and the pair $\{1,2,2,3\}$, $\{1,3,3,5\}$.

Checks

For the non-standard four-faced pair, list the totals: the 16 equally likely sums are $2,4,4,6,\ 3,5,5,7,\ 3,5,5,7,\ 4,6,6,8$, giving totals $2,\ldots,8$ with frequencies $1,2,3,4,3,2,1$, exactly as for two standard dice. Putting $t = 1$ in any of the pgfs above gives 1, a quick test of each factorisation. A computer search over all pairs of four-faced dice with labels from 1 to 7 finds exactly the two pairs above, and the totals for the $1,2,2,3,3,4$ and $1,3,4,5,6,8$ dice match the standard distribution exactly (for instance $\mathrm{P}(\text{total} = 7) = \frac{6}{36}$ in both cases).

Common slips

- Forgetting the constant c: you need a reason why one die can't carry a factor 2 and the other a factor $\frac{1}{2}$. The lowest coefficient being a positive integer for both dice is that reason.

- Allowing $k = 0$ or $k = 2$. Then one die would have a face labelled 0, which isn't a positive integer.
- Leaving $(1 - t + t^2)$ factorised into complex linear factors. Over the reals it's irreducible, and that's the form we need for sharing the factors out.

Beyond STEP

The six-sided pair here are the *Sicherman dice*, and the same method shows they are the only non-standard pair of six-sided dice with positive labels. The argument rests on unique factorisation of polynomials, which you'll meet in a first course in rings and fields as the polynomial version of the fundamental theorem of arithmetic. Generating functions also turn many hard counting problems into algebra. For both ideas, see Allenby, *Rings, Fields and Groups*, and Grimmett and Welsh, *Probability: An Introduction*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.